DETERMINING OPEN CIRCUIT LOCATION (HS-CAN) [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)]

id100220000400

Caution

- **Perform the following malfunction diagnosis only when it is diagnosed with a open circuit by CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW. (See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)].)**
- **If the malfunctioning part is detected in the communication line, before disconnecting the related connector for inspection, press the connector in the connection direction to verify that there is no looseness or disconnection.**
- **When disconnecting the connector, verify that there is no damage, deformation, or corrosion of the connector terminals.**

1. Verify the CAN system-related module DTCs and the failed module on the M-MDS screen.
2. Apply the communication error DTC and the failed module to DTC output pattern and malfunctioning location, and select the possible cause for the diagnostic result and the reference for the inspection item. (See DTC Output Pattern And Malfunctioning Location.)

Note

- The open circuit location can be determined by the DTC indicated in the DTC output pattern and malfunctioning location chart. DTCs not listed in the chart are not used for the determination of the open circuit location.(See DTC Output Pattern And Malfunctioning Location.)

3. Inspect the possible cause and inspection item of the applicable malfunctioning part.
4. After repairs, return to CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW, and verify that the repairs have been completed.(See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)].)

DTC Output Pattern And Malfunctioning Location

Cross (×): Communication error-related DTC and failed module

M-MDS display		DTC output pattern and malfunctioning location																		
DTC output module	**DTC**																			
PCM (PCM)	U0101:00					×														
	U0121:00		×																	
	U0131:00																×			
	U0140:00									×										
	U0151:00														×					
	U0155:00																			×
	U0214:00																		×	
	U0235:00											×								
	U0298:00			×																
ABS*1 (DSC HU/CM)	U0100:00	×																		
	U0101:00					×														
	U0131:00																×			
	U0151:00														×					
	U0155:00																			×
	U0235:00											×								
ABS*2 (ABS HU/CM)	U0100:00	×																		
	U0101:00					×														
	U0151:00														×					
	U0155:00																			×
DCDC*3 (DC-DC converter (i-ELOOP))	U0100:00	×																		
	U0140:00									×										
	U0151:00														×					
TCM*4 (TCM)	U0100:00	×			×															
	U0121:00		×		×															
	U0131:00																×			
	U0155:00																			×

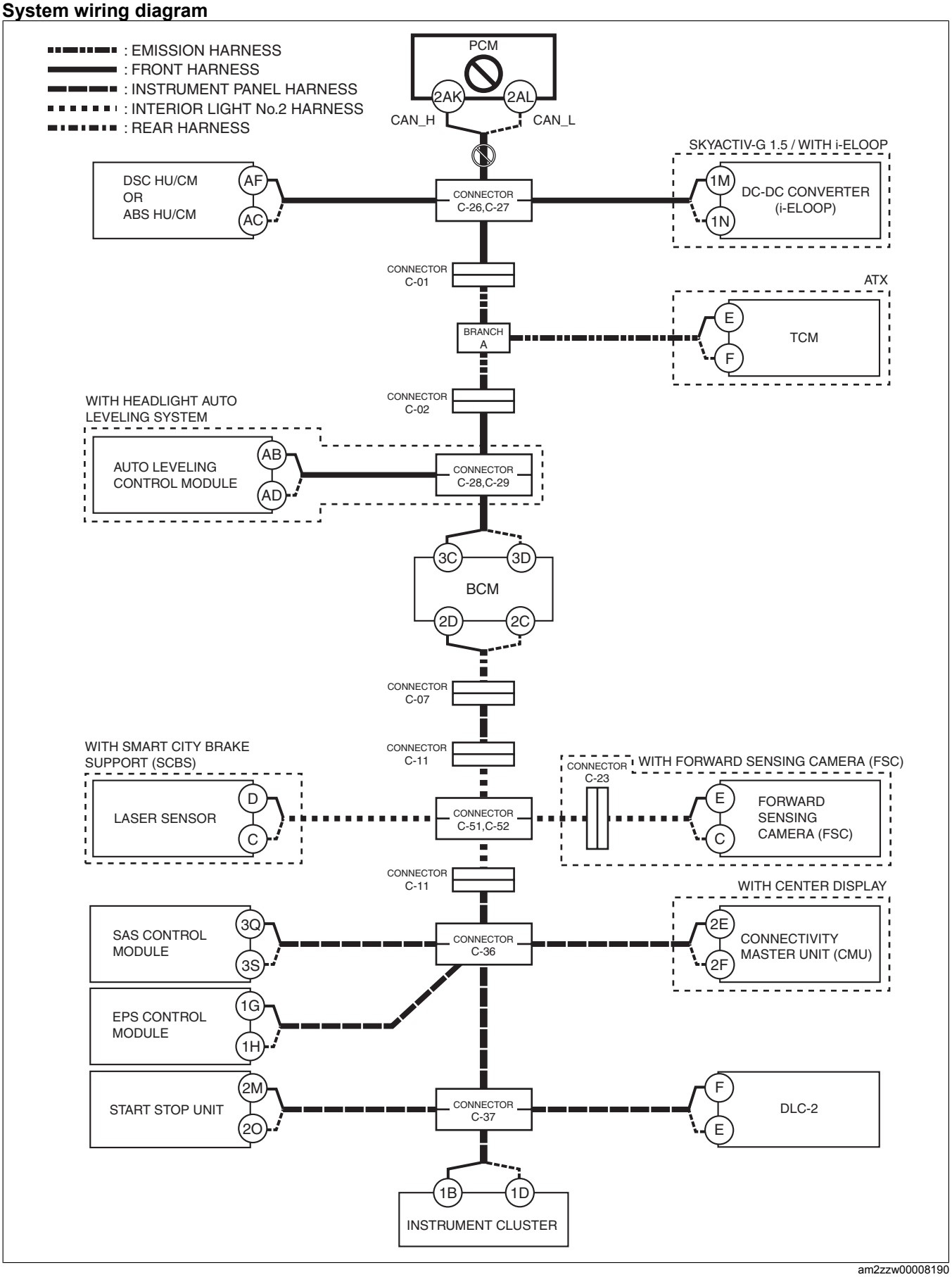
M-MDS display		DTC output pattern and malfunctioning location																		
DTC output module	DTC																			
ALM*5 (Auto leveling control module)	U0100:00	×			×		×													
	U0140:00									×										
	U0155:00																			×
BCM (BCM)	U0100:00	×			×		×		×											
	U0101:00					×	×		×											
	U0121:00		×		×		×		×											
	U0151:00														×					
	U0155:00																			×
	U0214:00																		×	
	U023A:00												×							
SCBS*6 (Laser sensor)	U0100:00	×			×		×		×		×									
	U0121:00		×		×		×		×		×									
	U0131:00																×			
	U0155:00																			×
FSC*7 (FORWARD SENSING CAMERA (FSC))	U0100:00	×			×		×		×		×									
	U0121:00		×		×		×		×		×									
	U0131:00																×			
	U0140:00									×	×									
	U0155:00																			×
	U0214:00																		×	
RCM (SAS control module)	U0100:00	×			×		×		×		×			×						
	U0101:00					×	×		×		×			×						
	U0131:00																×			
	U0155:00																			×
CMU*8 (Connectivity master unit (CMU))	U0100:00	×			×		×		×		×			×						
	U0101:00					×	×		×		×			×						
	U0121:00		×		×		×		×		×			×						
	U0131:00																×			
	U0140:00									×	×			×						
	U0151:00														×					
	U0155:00																			×
	U0182:00							×	×		×			×						
	U0214:00																		×	
	U0235:00											×		×						
	U023A:00												×	×						
EPS (EPS control module)	U0100:00	×			×		×		×		×			×						
	U0121:00		×		×		×		×		×			×						
	U0155:00																			×
	U0214:00																		×	
SSU (Start stop unit)	U0028:87									×	×			×				×		
	U0100:00	×			×		×		×		×			×				×		
	U0101:00					×	×		×		×			×				×		
	U0121:00		×		×		×		×		×			×				×		
	U0121:87		×		×		×		×		×			×				×		
	U0131:00																×	×		
	U0140:00									×	×			×				×		
	U0146:00									×	×			×				×		
	U0151:00														×			×		
	U0155:00																			×

M-MDS display		DTC output pattern and malfunctioning location																		
DTC output module	**DTC**																			
IC (Instrument cluster)	U0100:00	×			×		×		×		×			×				×		
	U0101:00					×	×		×		×			×				×		
	U0121:00		×		×		×		×		×			×				×		
	U0131:00																×	×		
	U0140:00									×	×			×				×		
	U0151:00														×			×		
	U0156:00															×		×		
	U0182:00							×	×		×			×				×		
	U0214:00																		×	
	U0235:00											×		×				×		
	U023A:00												×	×				×		
M-MDS display module		[Fail] display pattern																		
PCM		×			×		×		×		×			×				×		
ABS*1			×		×		×		×		×			×				×		
ABS*2			×		×		×		×		×			×				×		
DCDC*3				×	×		×		×		×			×				×		
TCM*4						×	×		×		×			×				×		
ALM*5								×	×		×			×				×		
BCM										×	×			×				×		
SCBS*6												×		×				×		
FSC*7													×	×				×		
RCM															×			×		
CMU*8																×		×		
EPS																	×	×		
SSU																			×	
IC																				×
Diagnostic result																				
Possible cause and inspection item		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S

*1 : With DSC system
*2 : Without DSC system
*3 : With i-ELOOP
*4 : ATX
*5 : With headlight auto leveling system
*6 : With smart city brake support (SCBS)
*7 : With forward sensing camera (FSC)
*8 : With center display

A
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- PCM power supply voltage or body ground malfunction
- Open circuit in wiring harness between PCM and connectors C-26,C-27
- Connectors C-26,C-27 malfunction
- PCM malfunction

System wiring diagram

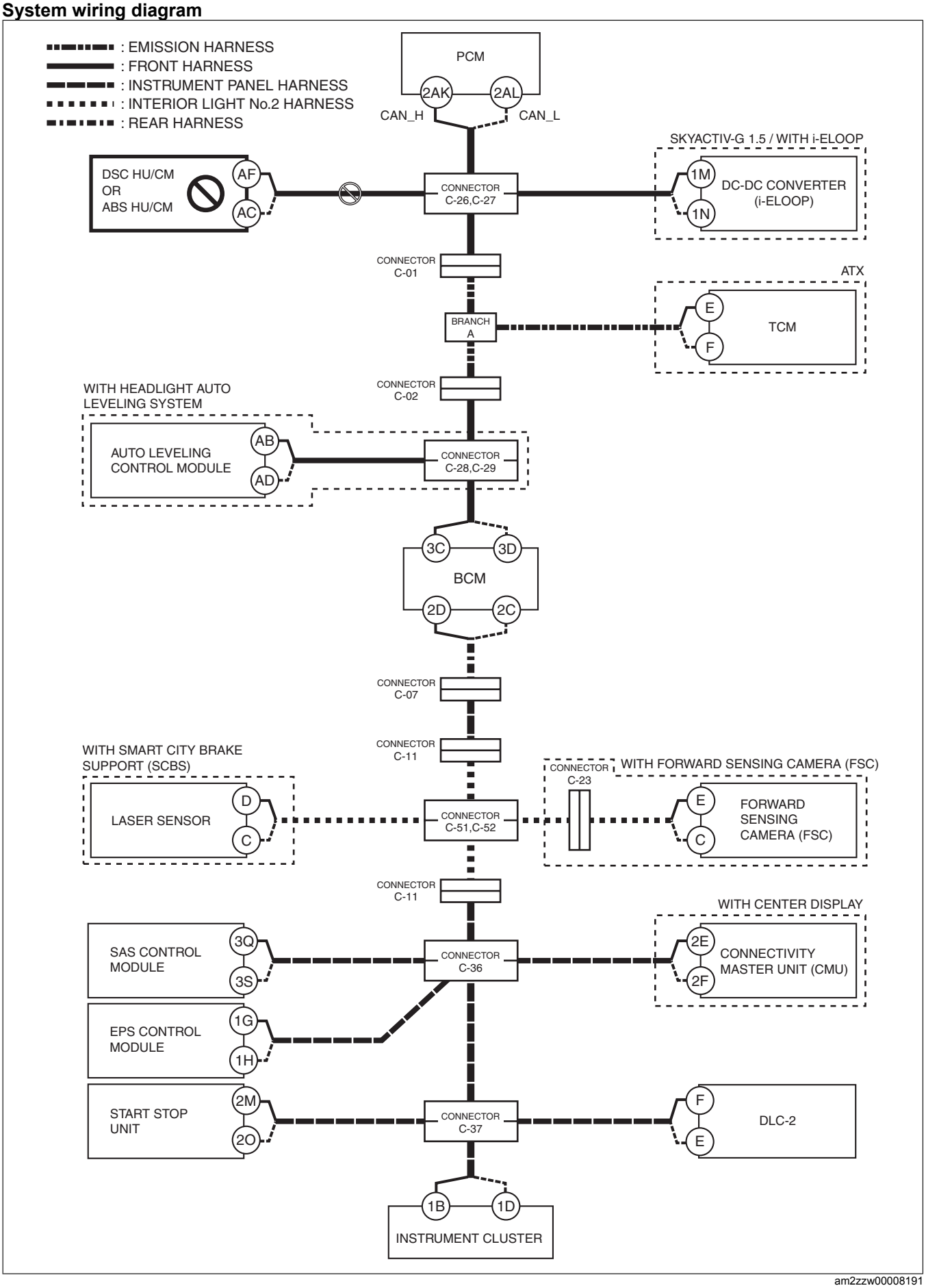


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Inspection item
- PCM power supply voltage or body ground malfunction
- PCM body ground related wiring harness
- PCM connector
- Connector C-26,C-27
- Wiring harness between PCM terminal 2AK and connector C-26
- Wiring harness between PCM terminal 2AL and connector C-27
- PCM

B
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- DSC HU/CM or ABS HU/CM power supply voltage or body ground malfunction
- Open circuit in wiring harness between DSC HU/CM or ABS HU/CM and connectors C-26,C-27
- Connectors C-26,C-27 malfunction
- DSC HU/CM or ABS HU/CM malfunction

System wiring diagram



am2zzw00008191

Inspection item

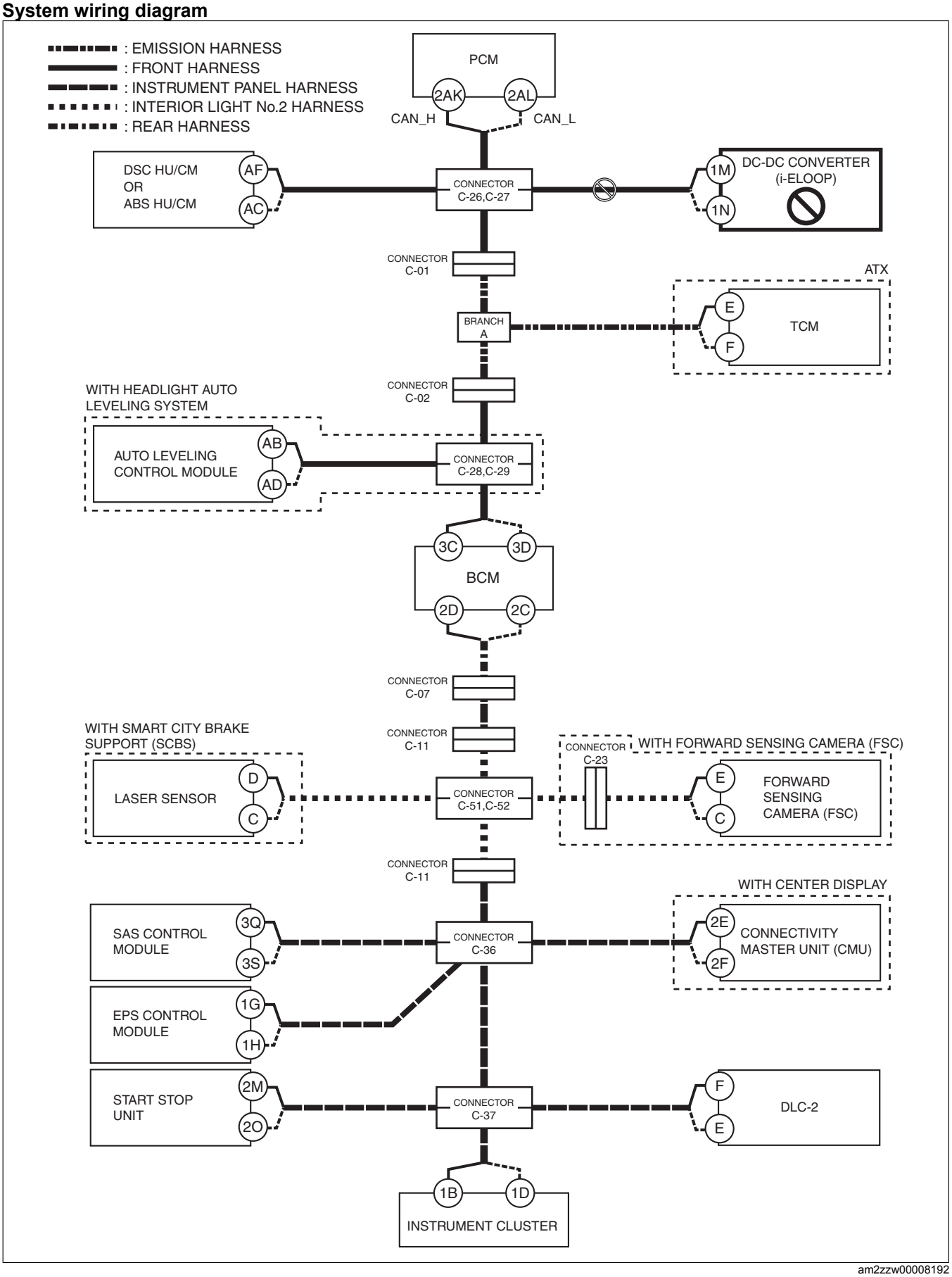
- DSC HU/CM or ABS HU/CM power supply voltage or body ground malfunction
- DSC HU/CM or ABS HU/CM body ground related wiring harness
- DSC HU/CM or ABS HU/CM connector
- Connectors C-26,C-27
- Wiring harness between DSC HU/CM or ABS HU/CM terminal AF and connector C-26
- Wiring harness between DSC HU/CM or ABS HU/CM terminal AC and connector C-27
- DSC HU/CM or ABS HU/CM

C

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- DC-DC converter (i-ELOOP) power supply voltage or body ground malfunction
- Open circuit in wiring harness between DC-DC converter (i-ELOOP) and connectors C-26,C-27
- Connectors C-26,C-27 malfunction
- DC-DC converter (i-ELOOP) malfunction

System wiring diagram



am2zzw00008192

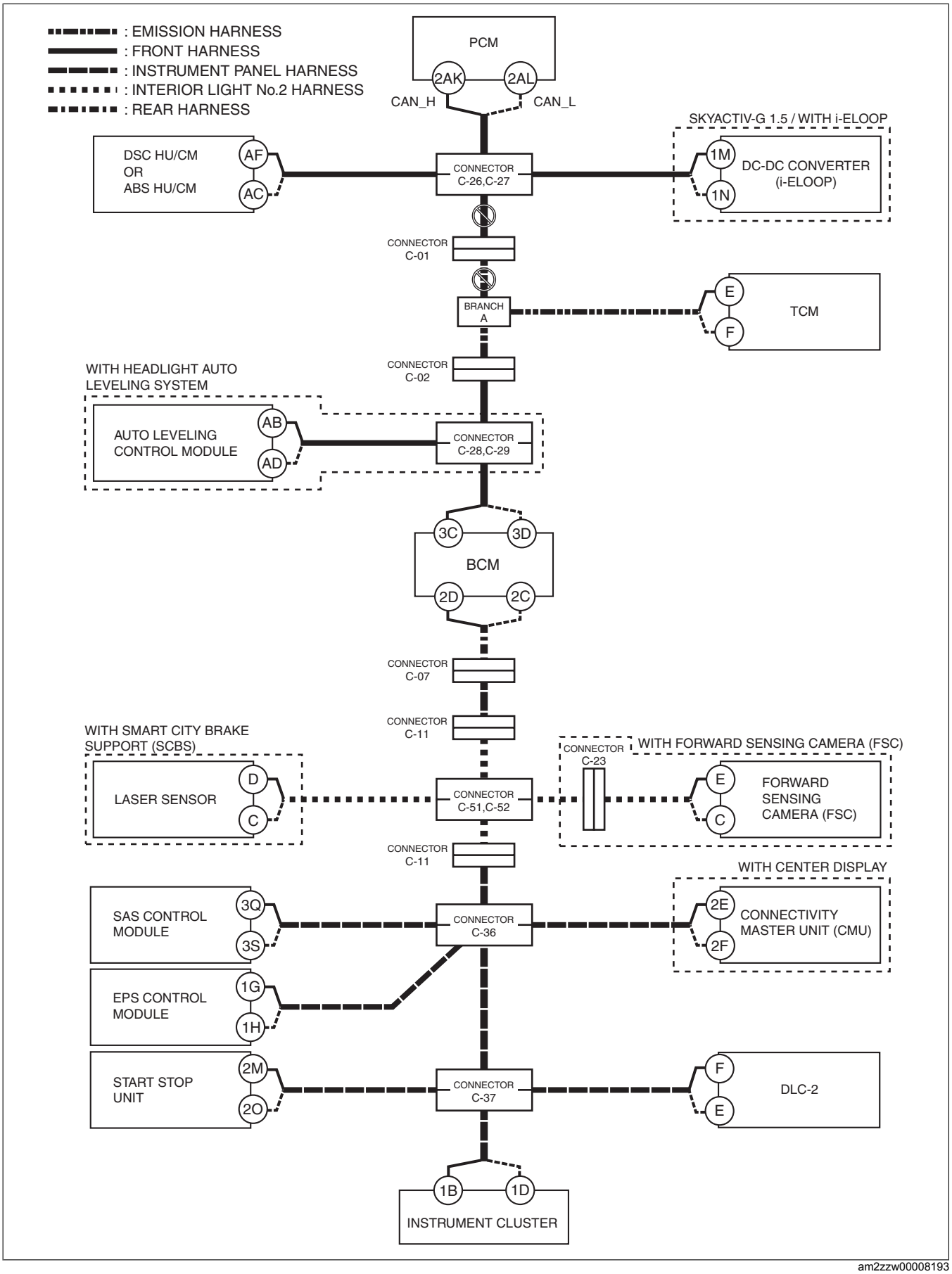
Inspection item

- DC-DC converter (i-ELOOP) power supply voltage or body ground malfunction
- DC-DC converter (i-ELOOP) body ground related wiring harness
- DC-DC converter (i-ELOOP) connector
- Connectors C-26,C-27
- Wiring harness between DC-DC converter (i-ELOOP) terminal 1M and connector C-26
- Wiring harness between DC-DC converter (i-ELOOP) terminal 1N and connector C-27
- DC-DC converter (i-ELOOP)

D
ATX
Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between connectors C-26,C-27 and connector C-01
- Open circuit in wiring harness between connector C-01 and branch A
- Connectors C-26,C-27 malfunction
- Connector C-01 malfunction

System wiring diagram



am2zzw00008193

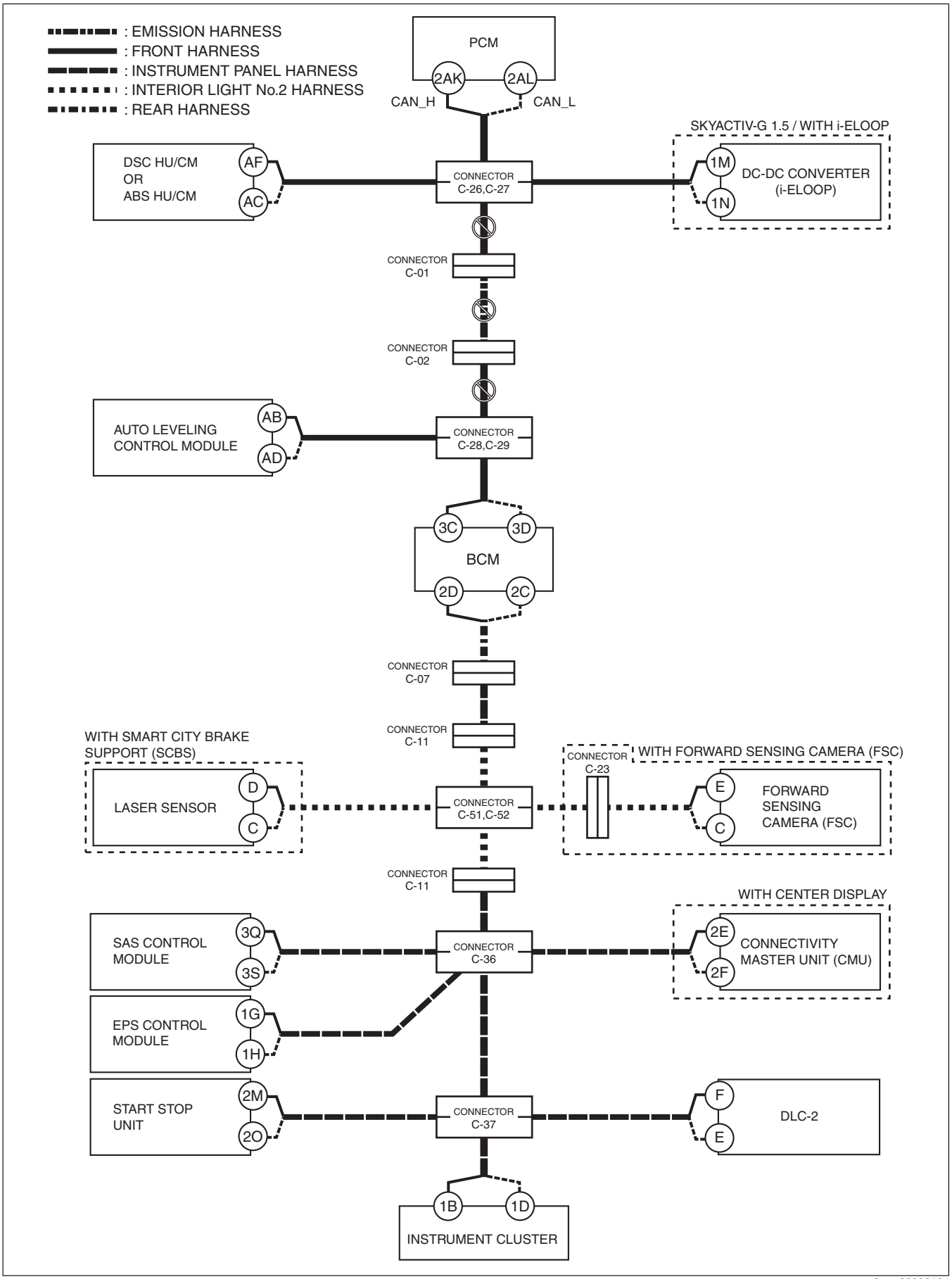
Inspection item

- Connectors C-26,C-27

- Connector C-01
- Wiring harness between connectors C-26,C-27 and connector C-01
- Wiring harness between connector C-01 and branch A

MTX (with headlight auto leveling system)
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between connectors C-26,C-27 and connector C-01
- Open circuit in wiring harness between connector C-01 and connector C-02
- Open circuit in wiring harness between connector C-02 and connectors C-28,C-29
- Connectors C-26,C-27 malfunction
- Connector C-01 malfunction
- Connector C-02 malfunction
- Connectors C-28,C-29 malfunction

System wiring diagram

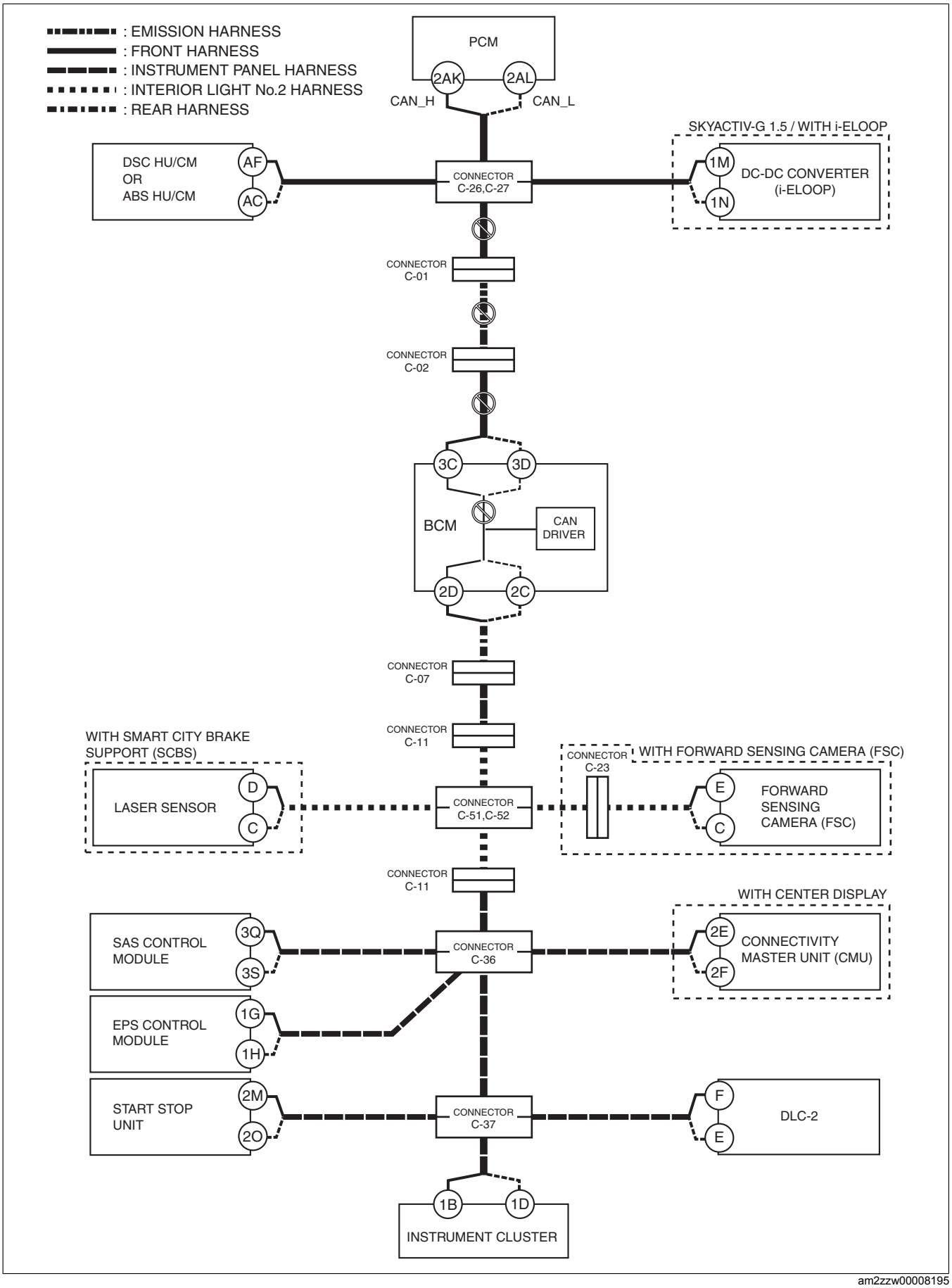


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Inspection item
- Connectors C-26,C-27
- Connector C-01
- Connector C-02
- Connectors C-28,C-29
- Wiring harness between connectors C-26,C-27 and connector C-01
- Wiring harness between connector C-01 and connector C-02
- Wiring harness between connector C-02 and connectors C-28,C-29

MTX (without headlight auto leveling system)
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between connectors C-26,C-27 and connector C-01
- Open circuit in wiring harness between connector C-01 and connector C-02
- Open circuit in wiring harness between connector C-02 and BCM
- Connectors C-26,C-27 malfunction
- Connector C-01 malfunction
- Connector C-02 malfunction
- CAN circuit in BCM malfunction

System wiring diagram



am2zzw00008195

Inspection item

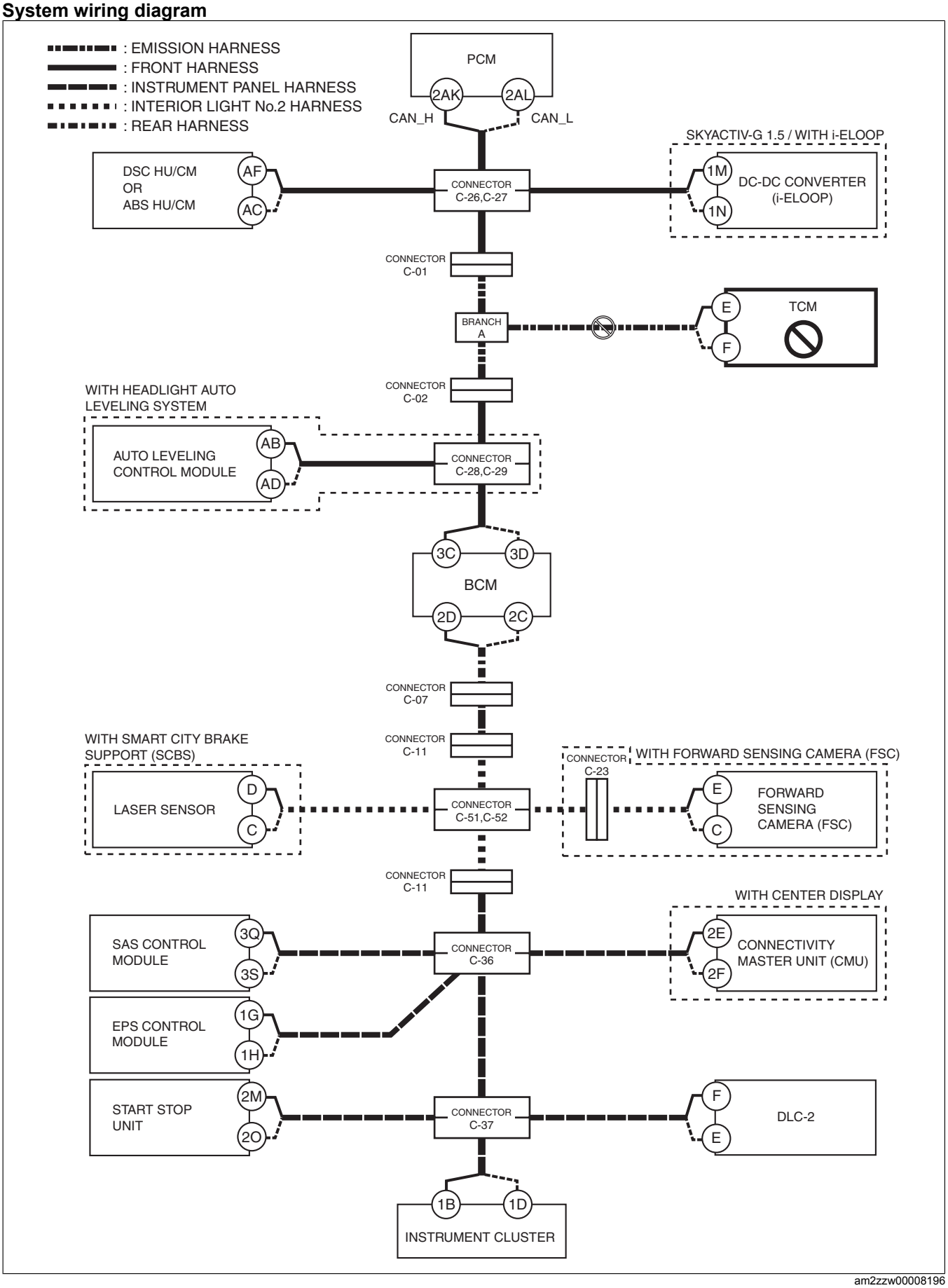
- Connectors C-26,C-27
- Connector C-01
- Connector C-02
- BCM connector
- Wiring harness between connectors C-26,C-27 and connector C-01
- Wiring harness between connector C-01 and connector C-02
- Wiring harness between BCM terminal 3C and connector C-02
- Wiring harness between BCM terminal 3D and connector C-02
- BCM
  - Between BCM terminal 3C and BCM terminal 2D
  - Between BCM terminal 3D and BCM terminal 2C

E

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- TCM power supply voltage or body ground malfunction
- Open circuit in wiring harness between TCM and branch A
- TCM malfunction

System wiring diagram



am2zzw00008196

Inspection item

- TCM power supply voltage or body ground malfunction
- TCM body ground related wiring harness
- TCM connector
- Wiring harness between TCM terminal E and branch A
- Wiring harness between TCM terminal F and branch A
- TCM

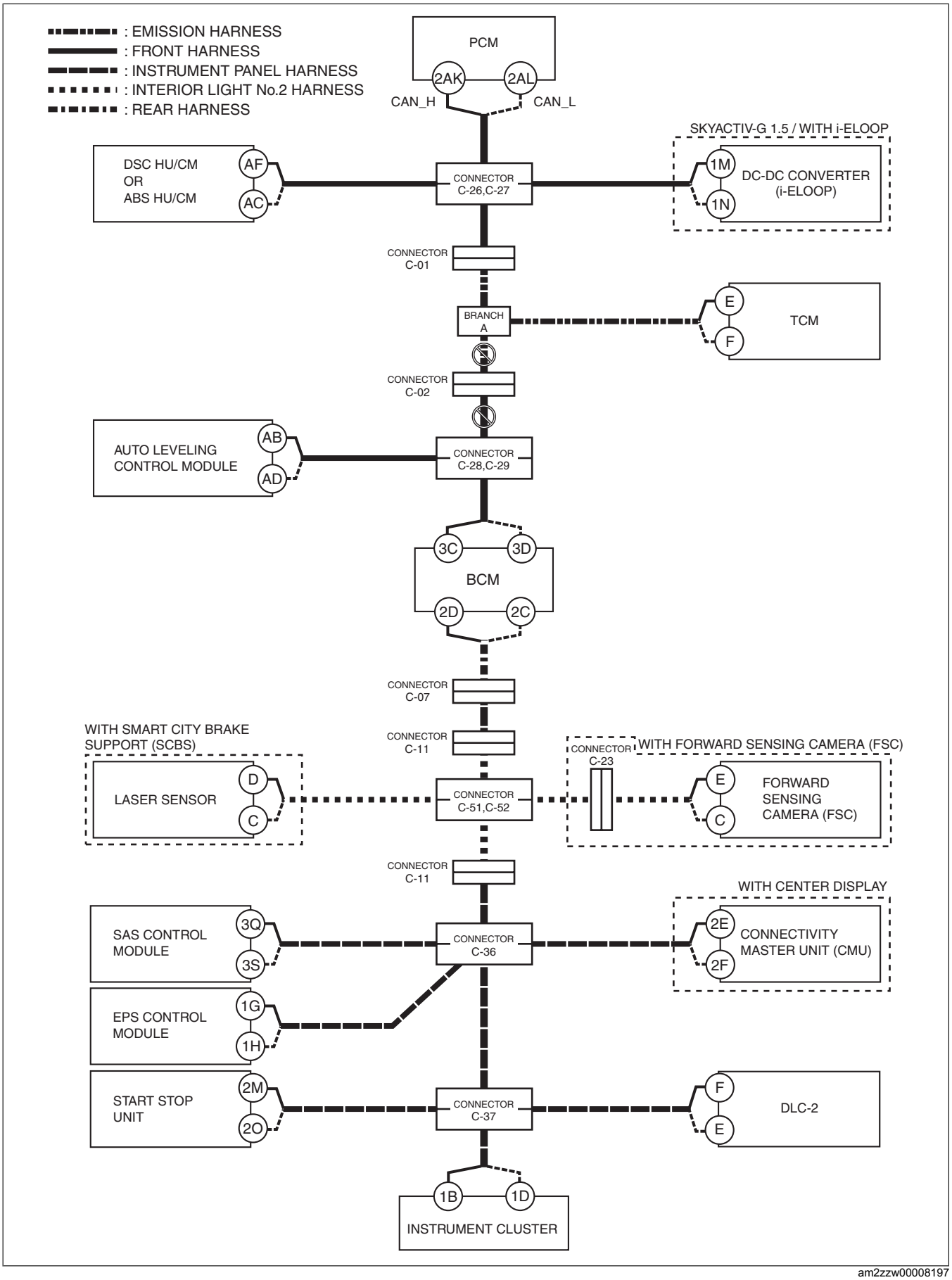
F

With headlight auto leveling system or i-ELOOP

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between branch A and connector C-02
- Open circuit in wiring harness between connector C-02 and connectors C-28,C-29
- Connector C-02 malfunction
- Connectors C-28,C-29 malfunction

System wiring diagram



am2zzw00008197

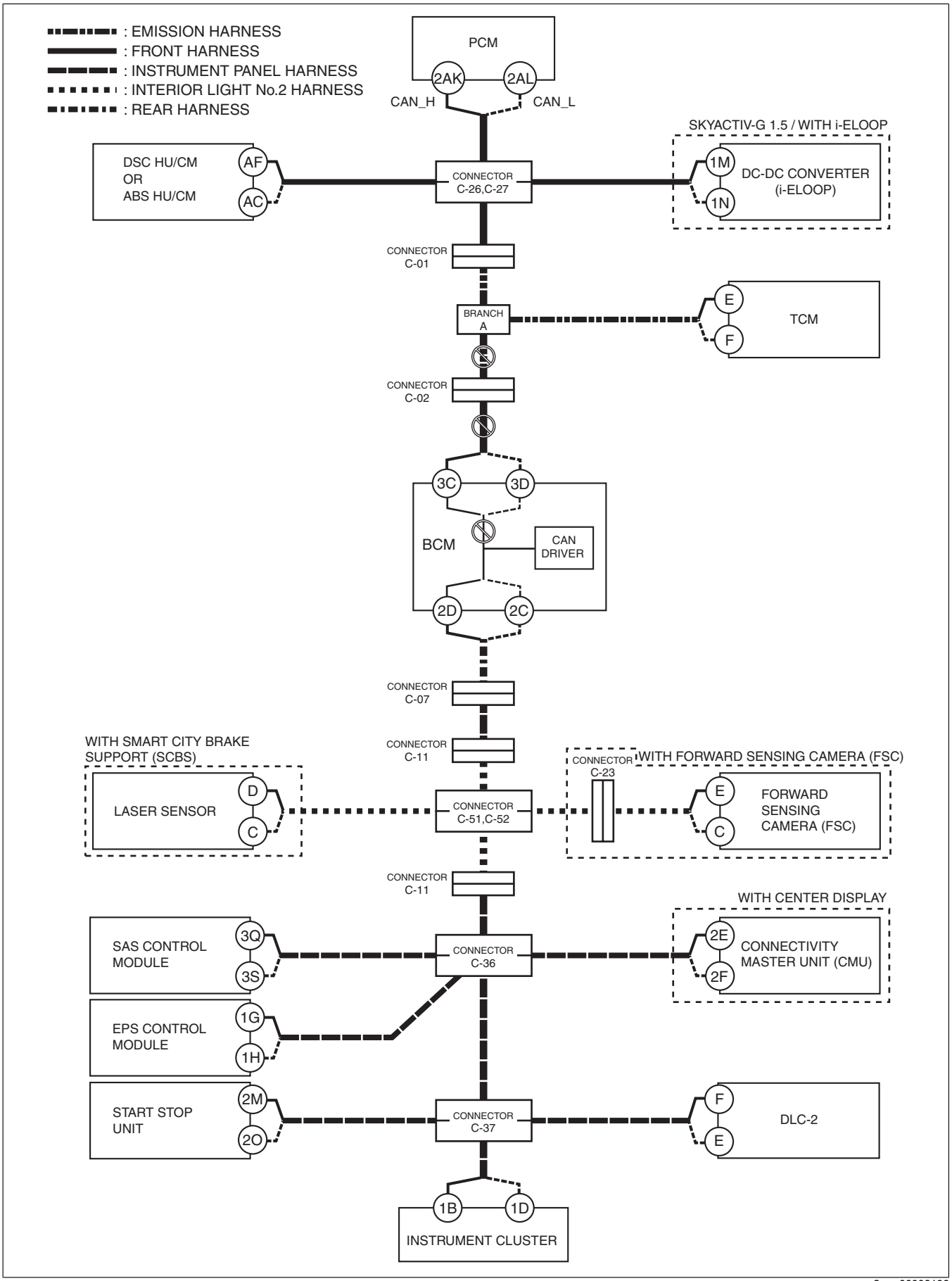
Inspection item

- Connector C-02

- Connectors C-28,C-29
- Wiring harness between branch A and connector C-02
- Wiring harness between connector C-02 and connectors C-28,C-29

Without headlight auto leveling system or i-ELOOP
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between branch A and connector C-02
- Open circuit in wiring harness between connector C-02 and BCM
- Connector C-02 malfunction
- CAN circuit in BCM malfunction

System wiring diagram

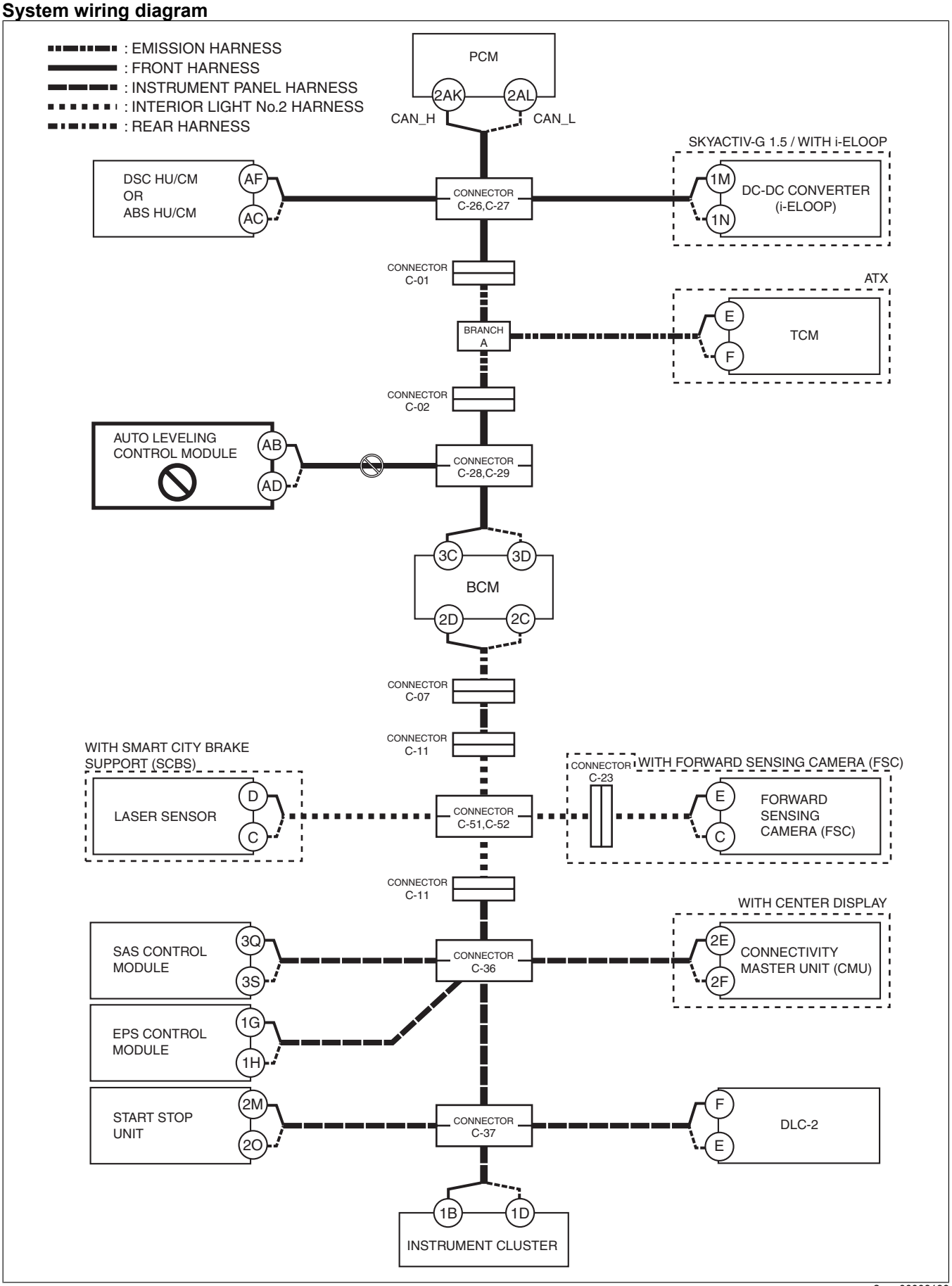


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Inspection item
- BCM connector
- Connector C-02
- Wiring harness between branch A and connector C-02
- Wiring harness between BCM terminal 3C and connector C-02
- Wiring harness between BCM terminal 3D and connector C-02
- BCM
  - — Between BCM terminal 3C and BCM terminal 2D
  - — Between BCM terminal 3D and BCM terminal 2C

G
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Auto leveling control module power supply voltage or body ground malfunction
- Open circuit in wiring harness between auto leveling control module and connectors C-28,C-29
- Connectors C-28,C-29 malfunction
- Auto leveling control module malfunction

System wiring diagram



am2zzw00008199

Inspection item

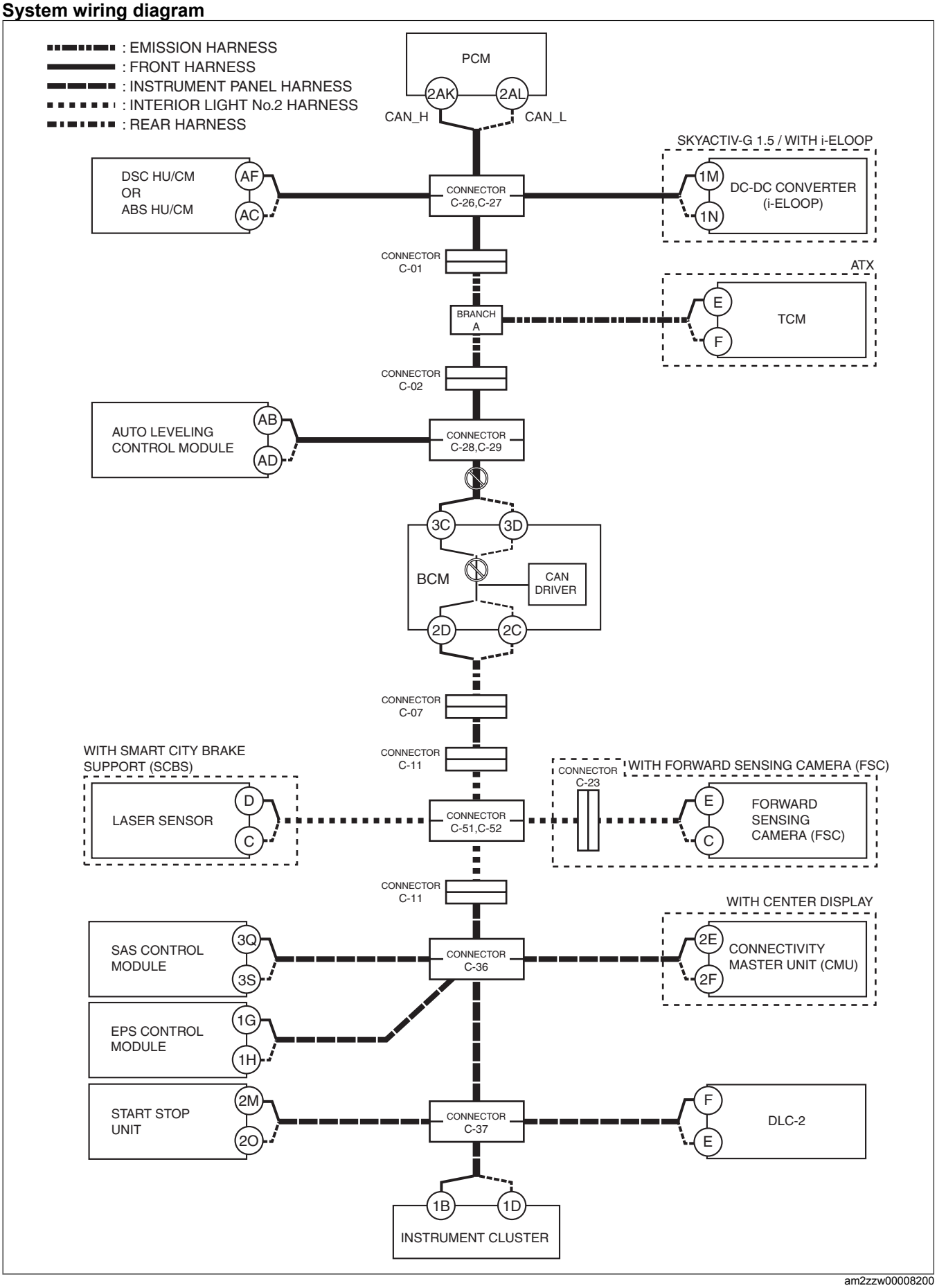
- Auto leveling control module power supply voltage or body ground malfunction
- Auto leveling control module body ground related wiring harness
- Auto leveling control module connector
- Connectors C-28,C-29
- Wiring harness between auto leveling control module terminal AB and connector C-28
- Wiring harness between auto leveling control module terminal AD and connector C-29
- Auto leveling control module

H

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between connectors C-28,C-29 and BCM
- Connectors C-28,C-29 malfunction
- CAN circuit in BCM malfunction

System wiring diagram



am2zzw00008200

Inspection item

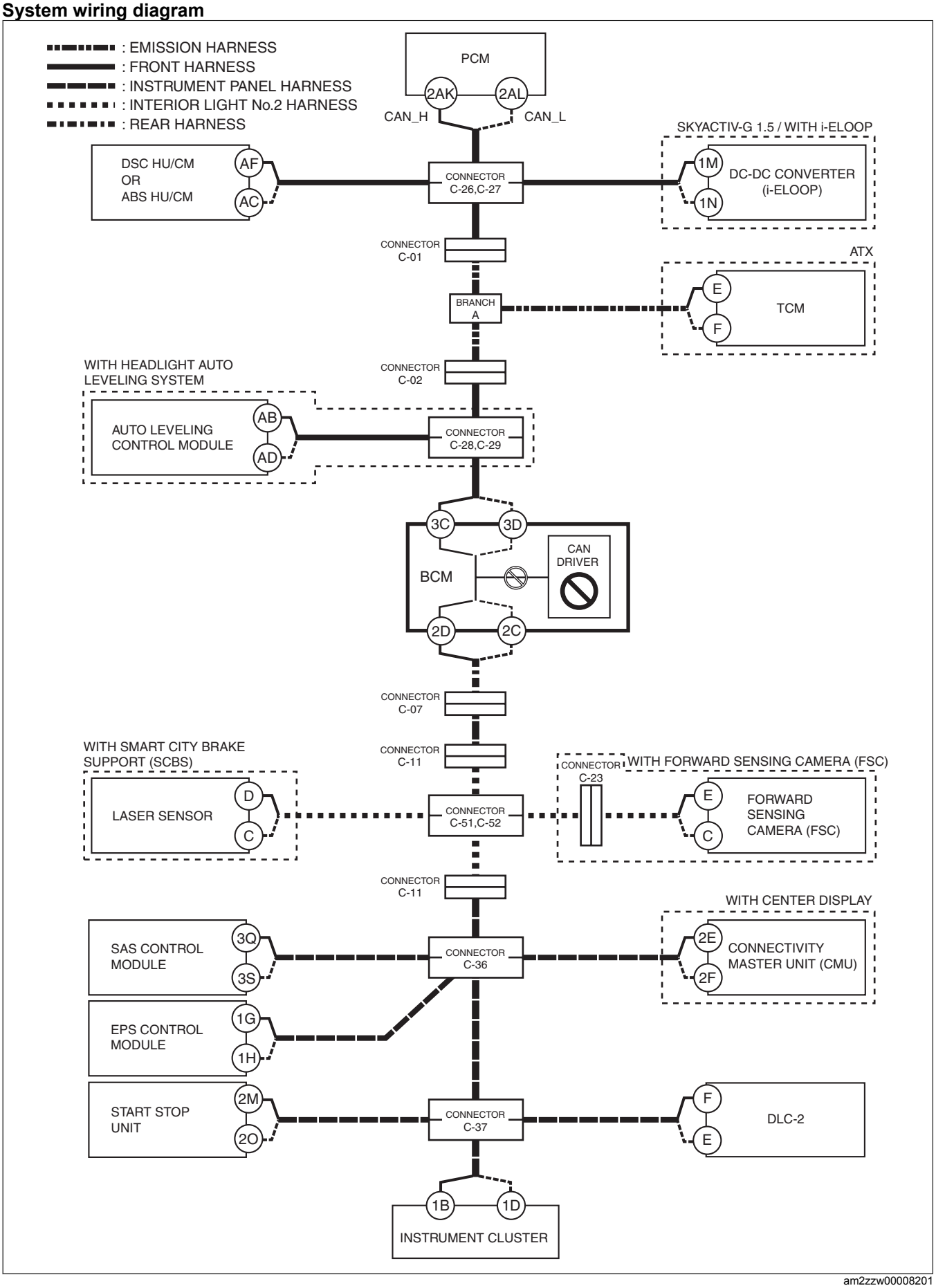
- Connectors C-28,C-29
- BCM connector
- Wiring harness between BCM terminal 3C and connector C-28
- Wiring harness between BCM terminal 3D and connector C-29
- BCM
  - — Between BCM terminal 3C and BCM terminal 2D
  - — Between BCM terminal 3D and BCM terminal 2C

I

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- BCM power supply voltage or body ground malfunction
- BCM malfunction

System wiring diagram

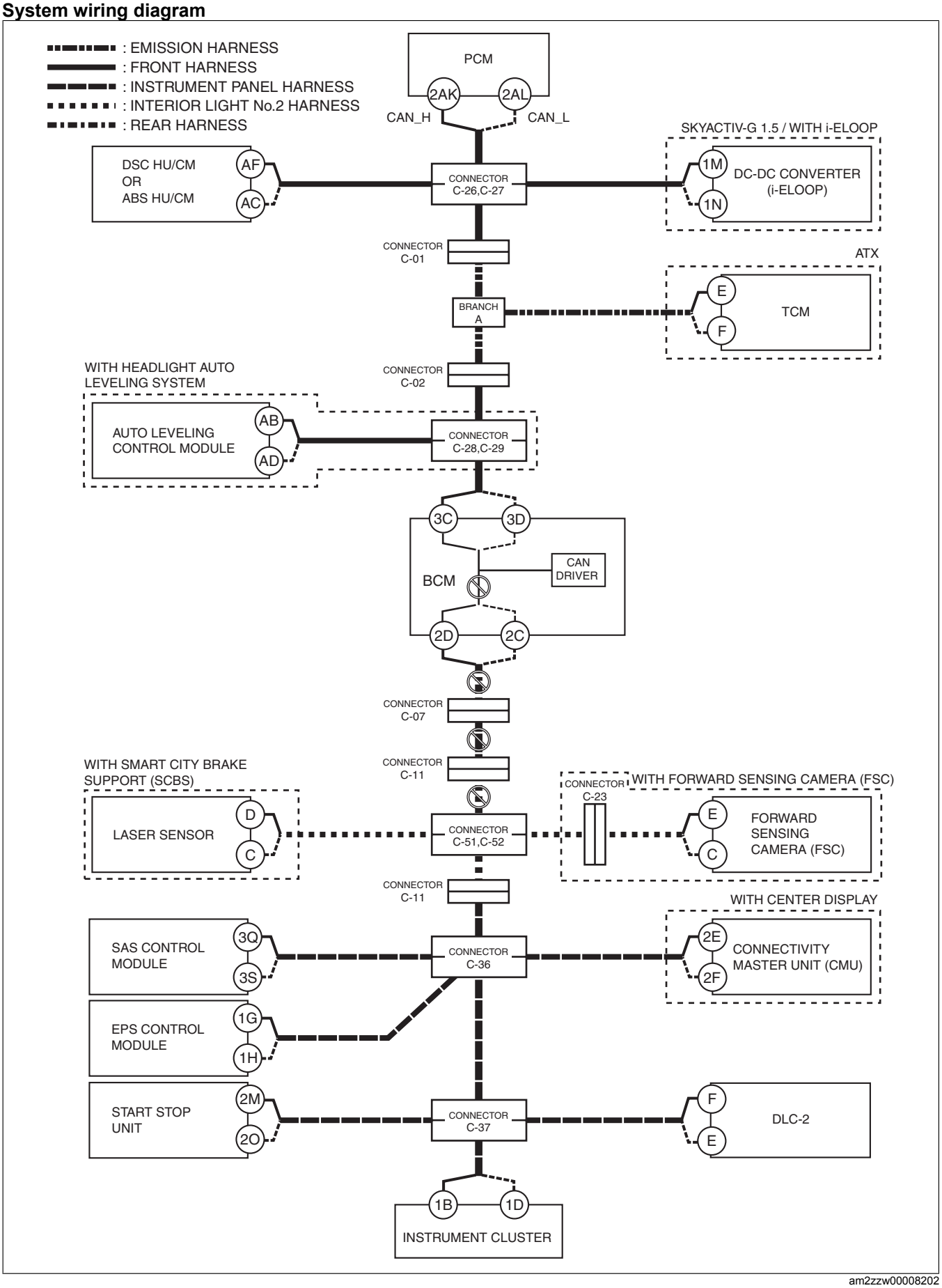


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Inspection item
- BCM power supply voltage or body ground malfunction
- BCM body ground related wiring harness
- BCM

J
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between BCM and connector C-07
- Open circuit in wiring harness between connector C-07 and connector C-11
- Open circuit in wiring harness between connector C-11 and connectors C-51,C-52
- Connector C-07 malfunction
- Connector C-11 malfunction
- Connectors C-51,C-52 malfunction
- CAN circuit in BCM malfunction

System wiring diagram



am2zzw00008202

Inspection item

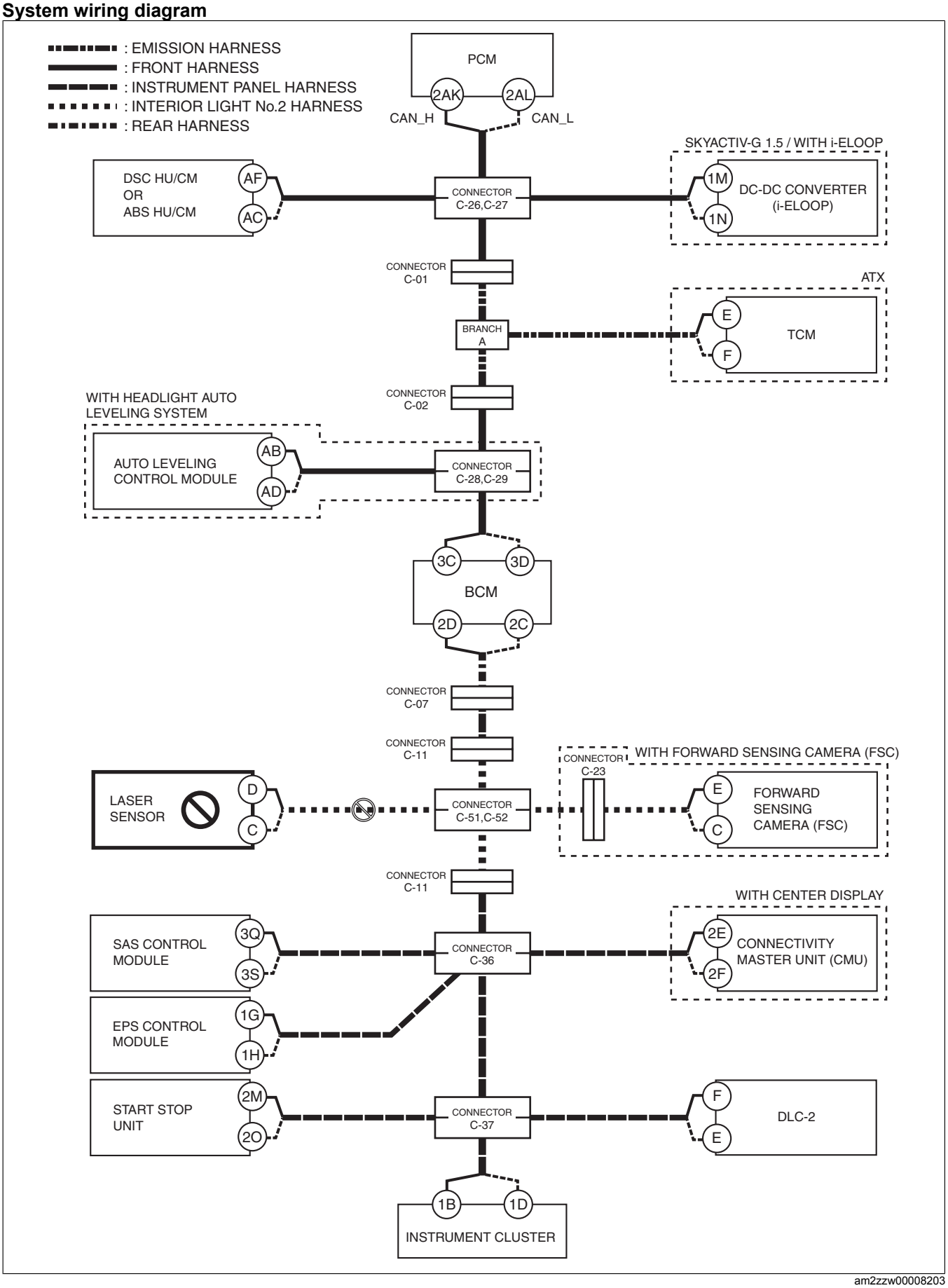
- Connector C-07
- Connector C-11
- Connectors C-51,C-52
- BCM connector
- Wiring harness between BCM terminal 2D and connector C-07
- Wiring harness between BCM terminal 2C and connector C-07
- Wiring harness between connector C-07 and connector C-11
- Wiring harness between connector C-11 and connectors C-51,C-52
- BCM
  - Between BCM terminal 3C and BCM terminal 2D
  - Between BCM terminal 3D and BCM terminal 2C

K

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Laser sensor power supply voltage or body ground malfunction
- Open circuit in wiring harness between laser sensor and connectors C-51,C-52
- Connectors C-51,C-52 malfunction
- Laser sensor malfunction

System wiring diagram



am2zzw00008203

Inspection item

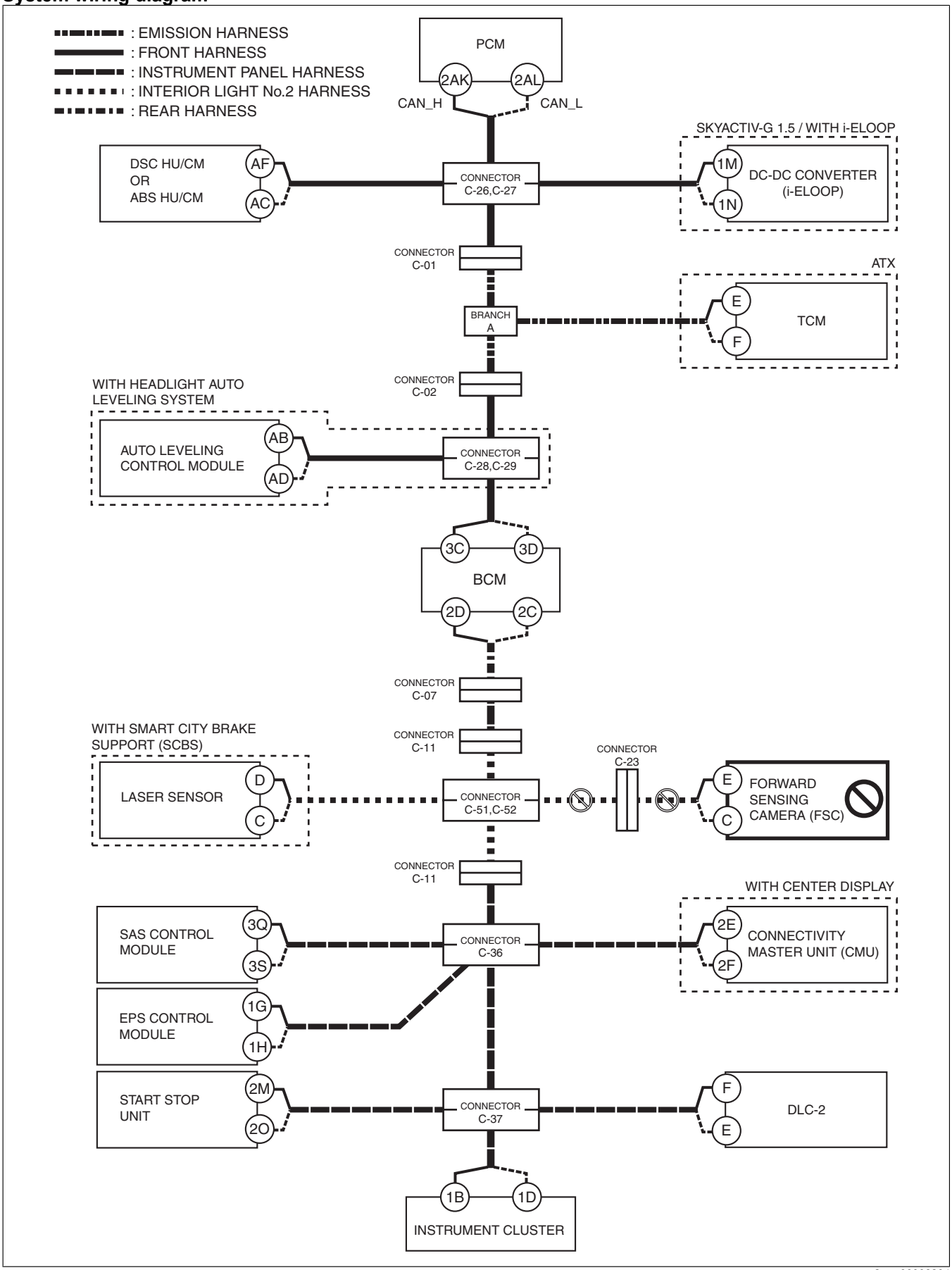
- Laser sensor power supply voltage or body ground malfunction
- Laser sensor body ground related wiring harness
- Laser sensor connector
- Connectors C-51,C-52
- Wiring harness between laser sensor terminal D and connector C-51
- Wiring harness between laser sensor terminal C and connector C-52
- Laser sensor

L

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Forward sensing camera (FSC) power supply voltage or body ground malfunction
- Open circuit in wiring harness between forward sensing camera (FSC) and connector C-23
- Open circuit in wiring harness between connector C-23 and connectors C-51,C-52
- Connector C-23 malfunction
- Connectors C-51,C-52 malfunction
- Forward sensing camera (FSC) malfunction

System wiring diagram



am2zzw00008204

Inspection item

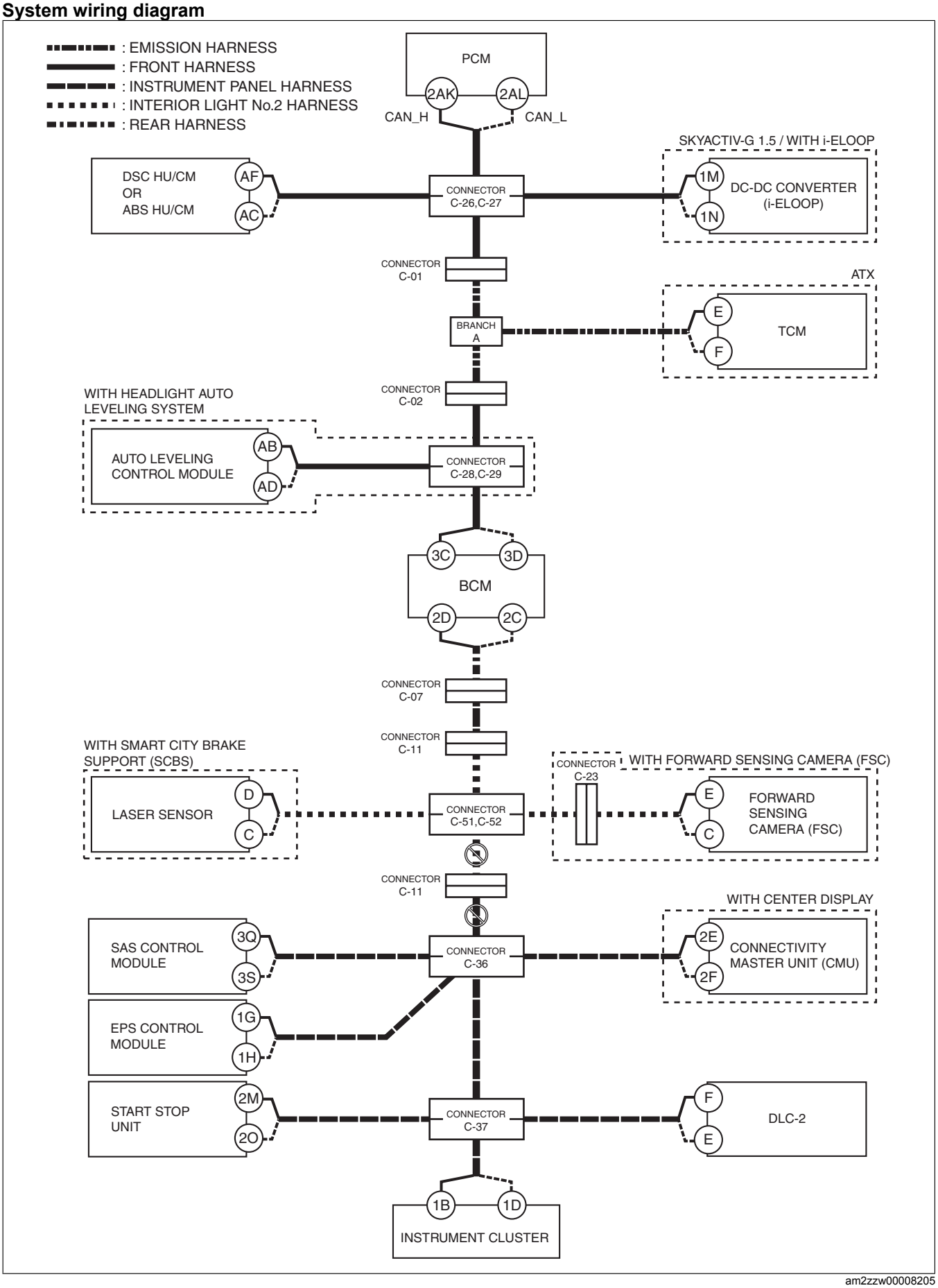
- Forward sensing camera (FSC) power supply voltage or body ground malfunction
- Forward sensing camera (FSC) body ground related wiring harness
- Forward sensing camera (FSC) connector
- Connectors C-51,C-52
- Connector C-23
- Wiring harness between forward sensing camera (FSC) terminal E and connector C-23
- Wiring harness between forward sensing camera (FSC) terminal C and connector C-23
- Wiring harness between connector C-23 and connectors C-51,C-52
- Forward sensing camera (FSC)

M

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between connectors C-51,C-52 and connector C-11
- Open circuit in wiring harness between connector C-11 and connector C-36
- Connectors C-51,C-52 malfunction
- Connector C-11 malfunction
- Connector C-36 malfunction

System wiring diagram

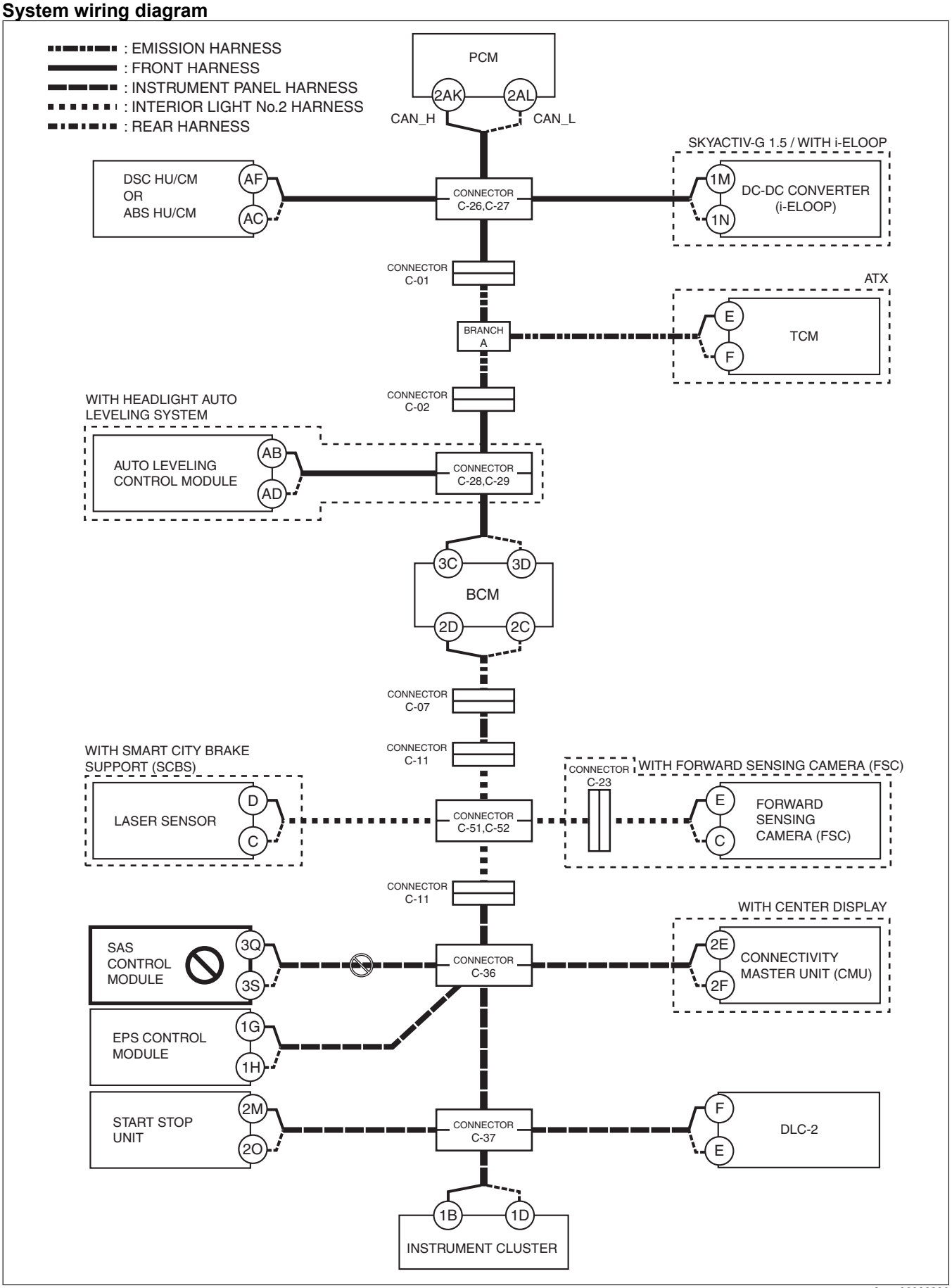


am2zzw00008205

Inspection item
- Connectors C-51,C-52
- Connector C-11
- Connector C-36
- Wiring harness between connectors C-51,C-52 and connector C-11
- Wiring harness between connector C-11 and connector C-36

N
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- SAS control module power supply voltage or body ground malfunction
- Open circuit in wiring harness between SAS control module and connector C-36
- Connector C-36 malfunction
- SAS control module malfunction

System wiring diagram



am2zzw00008206

Inspection item

Warning

- **Handling the component parts of the SRS air bag system improperly can accidentally operate (deploy) the air bag module, which may seriously injure you. To prevent this, always read the SRS air bag system service warnings and cautions before handling the SRS air bag system components. (See AIR BAG SYSTEM SERVICE WARNINGS [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) (See AIR BAG SYSTEM SERVICE WARNINGS [STANDARD DEPLOYMENT CONTROL SYSTEM].) (See AIR BAG SYSTEM SERVICE CAUTIONS [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) (See AIR BAG SYSTEM SERVICE CAUTIONS [STANDARD DEPLOYMENT CONTROL SYSTEM].)**

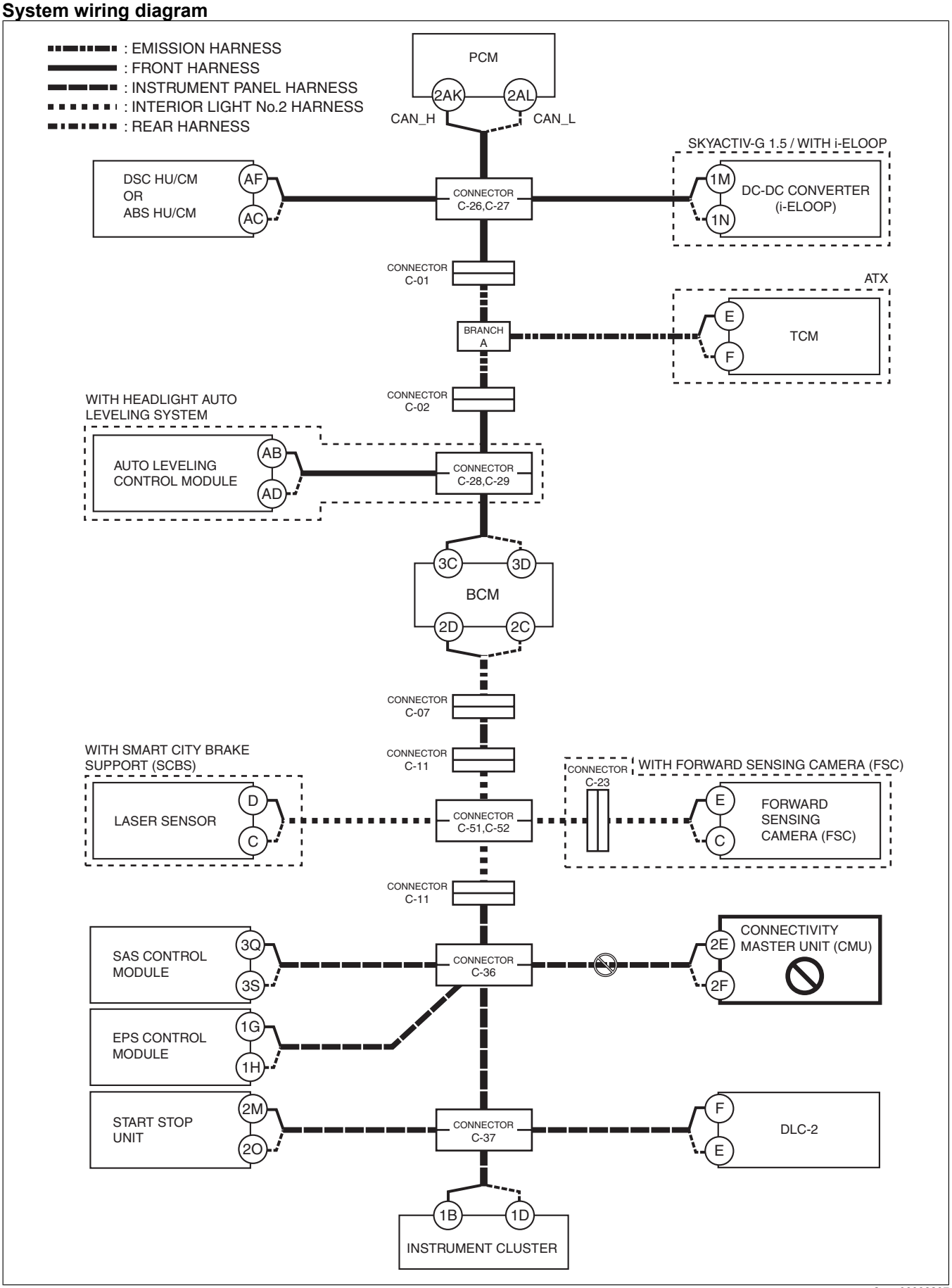
- SAS control module power supply voltage or body ground malfunction
- SAS control module body ground related wiring harness
- SAS control module connector
- Connector C-36
- Wiring harness between SAS control module terminal 3Q and connector C-36
- Wiring harness between SAS control module terminal 3S and connector C-36
- SAS control module

O

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Connectivity master unit (CMU) power supply voltage or body ground malfunction
- Wiring harness between connectivity master unit (CMU) and connector C-37
- Connector C-36 malfunction
- Connectivity master unit (CMU) malfunction

System wiring diagram



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Inspection item

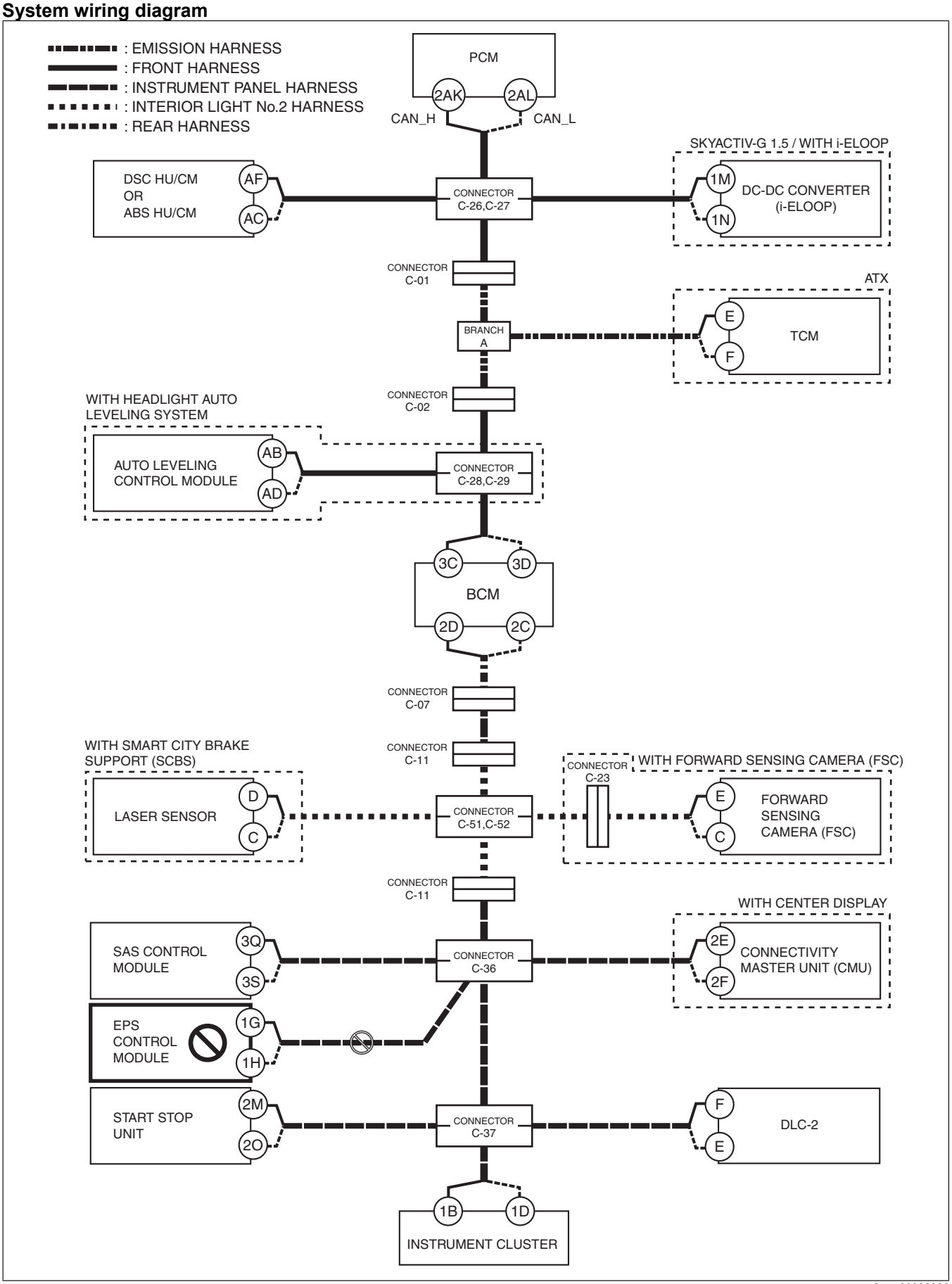
- Connectivity master unit (CMU) power supply voltage or body ground malfunction
- Connectivity master unit (CMU) body ground related wiring harness
- Connectivity master unit (CMU) connector
- Connector C-36
- Wiring harness between connectivity master unit (CMU) terminal 2E and connector C-36
- Wiring harness between connectivity master unit (CMU) terminal 2F and connector C-36
- Connectivity master unit (CMU)

P

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- EPS control module power supply voltage or body ground malfunction
- Open circuit in wiring harness between EPS control module and connector C-36
- Connector C-36 malfunction
- EPS control module malfunction

System wiring diagram

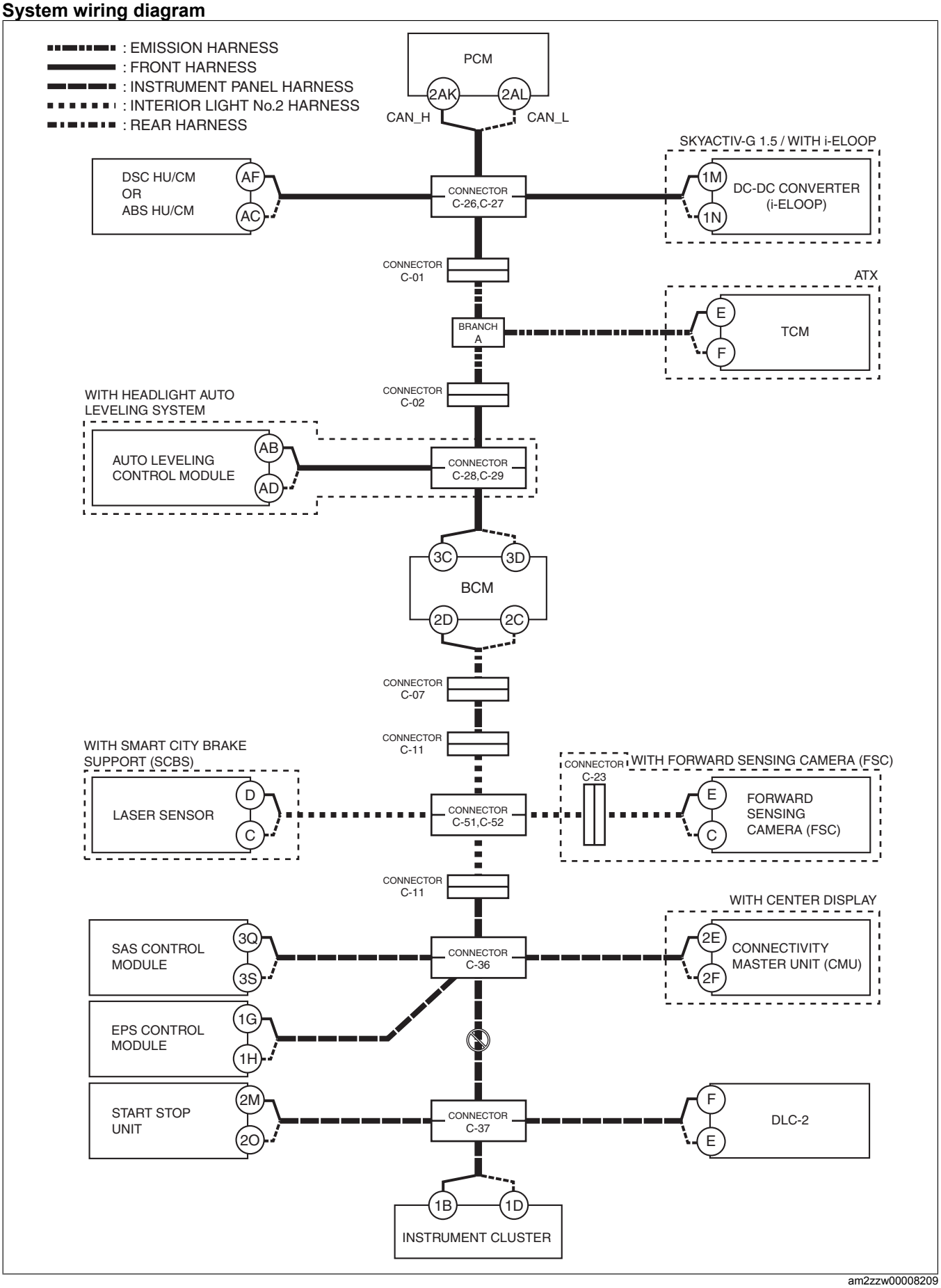


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Inspection item
- EPS control module power supply voltage or body ground malfunction
- EPS control module body ground related wiring harness
- EPS control module connector
- Connector C-36
- Wiring harness between EPS control module terminal 1G and connector C-36
- Wiring harness between EPS control module terminal 1H and connector C-36
- EPS control module

Q
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between connector C-36 and connector C-37
- Connector C-36 malfunction
- Connector C-37 malfunction

System wiring diagram

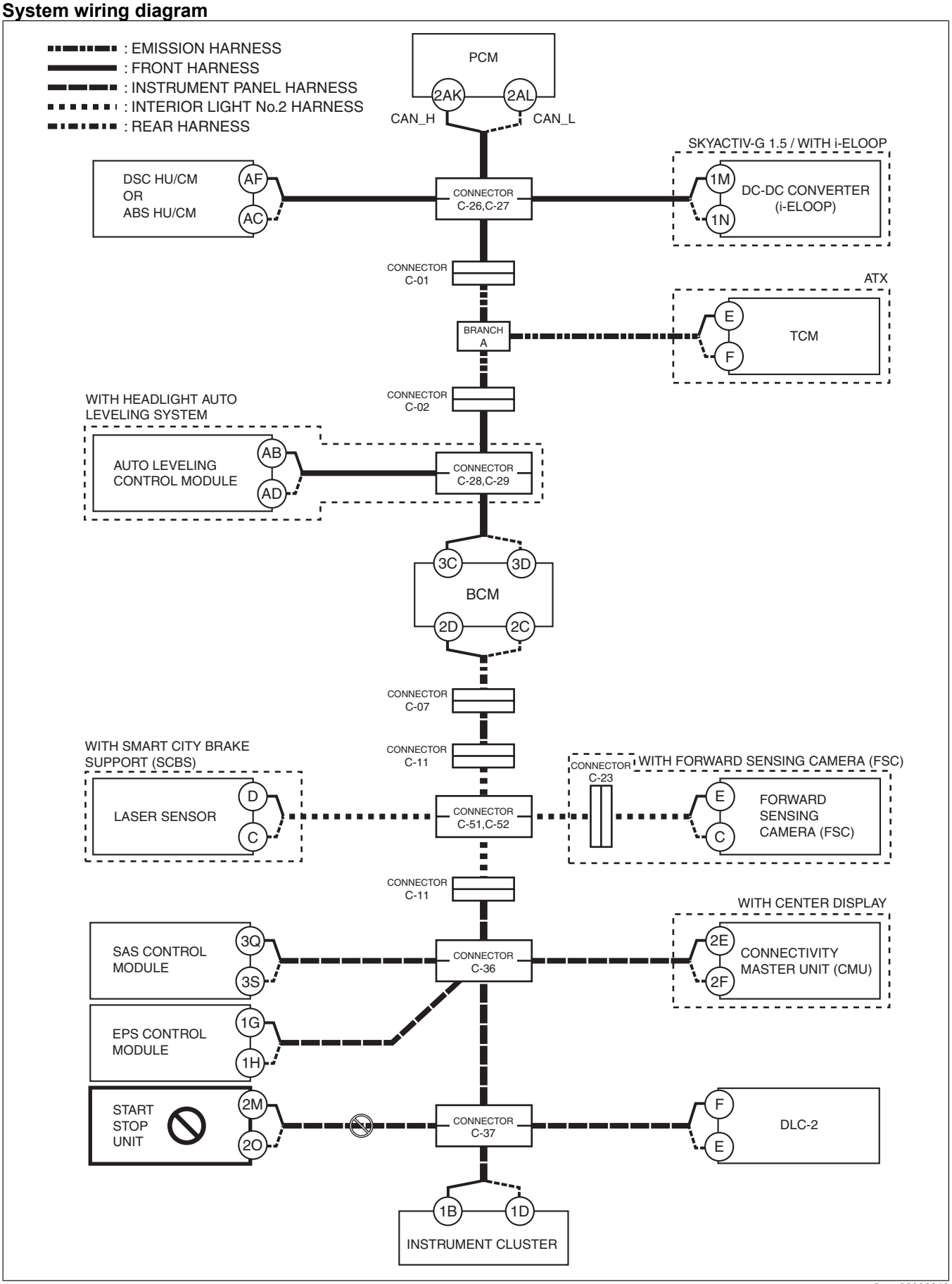


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Inspection item
- Connector C-36
- Connector C-37
- Wiring harness between connector C-36 and connector C-37

R
Possible cause
- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Start stop unit power supply voltage or body ground malfunction
- Open circuit in wiring harness between start stop unit and connector C-37
- Connector C-37 malfunction
- Start stop unit malfunction

System wiring diagram



am2zzw00008210

Inspection item

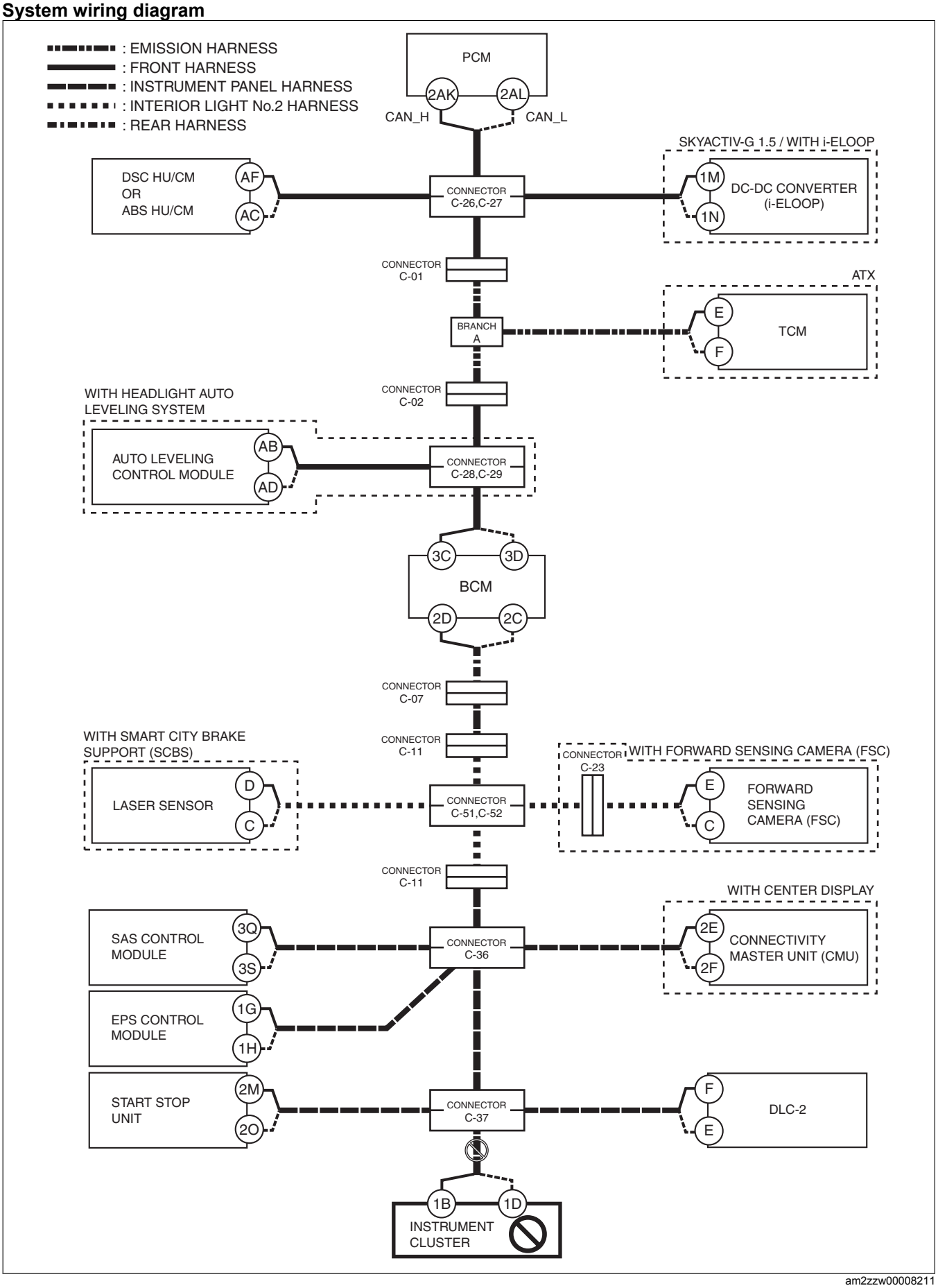
- Start stop unit power supply voltage or body ground malfunction
- Start stop unit body ground related wiring harness
- Start stop unit connector
- Connector C-37
- Wiring harness between start stop unit terminal 2M and connector C-37
- Wiring harness between start stop unit terminal 2O and connector C-37
- Start stop unit

S

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Instrument cluster power supply voltage or body ground malfunction
- Open circuit in wiring harness between instrument cluster and connector C-37
- Connector C-37 malfunction
- Instrument cluster malfunction

System wiring diagram



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Inspection item

- Instrument cluster power supply voltage or body ground malfunction
- Instrument cluster body ground related wiring harness
- Instrument cluster connector
- Connector C-37
- Wiring harness between instrument cluster terminal 1B and connector C-37
- Wiring harness between instrument cluster terminal 1D and connector C-37
- Instrument cluster